STAFF REVIEW · H446 146 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Use sequence information to reassemble the byte stream correctly.

2. The internet layer.

3. A queue.

4. 0.

I Do — source worked steps

1. Fake login email — Identify the facts and make an initial prediction.

2. Credentials submitted — Explain the mechanism or trace the intermediate stage.

3. Account access at risk — Phishing tricks the pupil into visiting an attacker-controlled form. The secret is disclosed, then the attacker may attempt unauthorised account access. Awareness and phishing-resistant MFA help, while a firewall alone does not prevent users entering secrets into fake sites.

We Do — guided CHECK

1. Botnet generates traffic
Identify the facts and make an initial prediction.
Server/link overloaded
Explain the mechanism or trace the intermediate stage.
Mitigate upstream
A distributed denial-of-service (DDoS) attack aims to exhaust bandwidth or service resources. Upstream rate-limiting, DDoS protection and filtering may reduce impact; encrypting the website does not stop resource saturation.

You Do — source mark schemes

1. Phishing deceives a user into divulging information or performing an action [3]; malware is malicious software with a damaging/unauthorised function, e.g. ransomware [3].

2. Confidentiality breached [2]; e.g. MFA/least privilege/encrypted connections, secure authentication, anti-phishing education or monitoring linked to attack [2+2].

3. DDoS upstream mitigation [2], account containment/MFA/reset [2], differing threat mechanism [2], defence-in-depth explanation [2].

Exit Ticket — source answer

1. False. Encryption protects traffic confidentiality/integrity in transit, not server capacity.